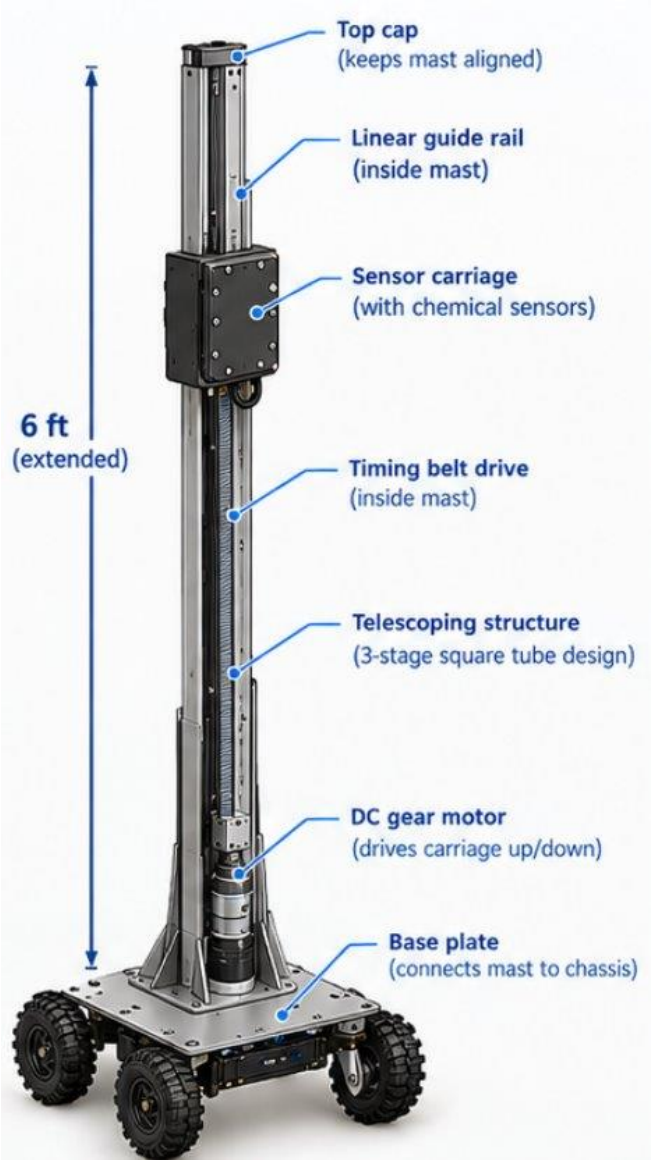
Project Sentry

Team 4: Jenna Andrews, Emily Johns,

Owen Swartout, Tu Tran, Lucas Witthauer



Sentry

Team Sentry: Lucas Witthauer (EE), Emily Johns (ME), Jenna Andrews (GE), Owen Swartout (EE), Tu Tran (ME)

Objective:

- Create a safe and easy way to prevent industrial workers from being exposed to dangerous chemicals.
- Providing real-time chemical data information improves hazard awareness and reduces the need for workers to enter potentially dangerous areas just to perform air-quality checks.
- Potential applications include industrial facilities, manufacturing plants, and laboratories.

Concept:

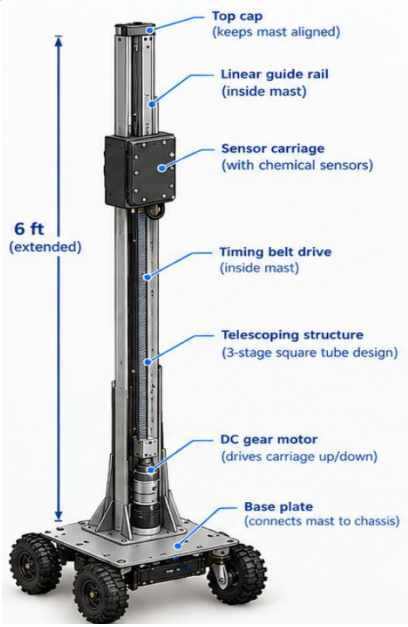
Compact, remote scout robot that enters possibly hazardous areas ahead of workers, samples the air and alerts the operator wirelessly when chemical levels become unsafe.

•Gas sensing: A vertically extendable arm with sensors attached on a movable carriage.

•Detection alerts: Operator station notification within 3 seconds of threshold crossing

•Range: 30 meters wireless link, driven from a safe distance

•Runtime: 2 hours on battery



Approach:

Develop a remotely operated chemical detection rover by:

- Developing a rover that can be controlled wirelessly.
- Design an extending arm that positions chemical sensors at different heights.
- Integrating appropriate chemical sensors with a microcontroller to test and transmit data to the operator in real time.
- Developing a power distribution system that reliably supplies the motors, arm actuator, sensors, and control electronics.

Analyses and RRP:

Analyses

- Robot car balance and structural integrity
- Vertical arm balance and structural integrity
- Electrical power budget and estimated power consumption

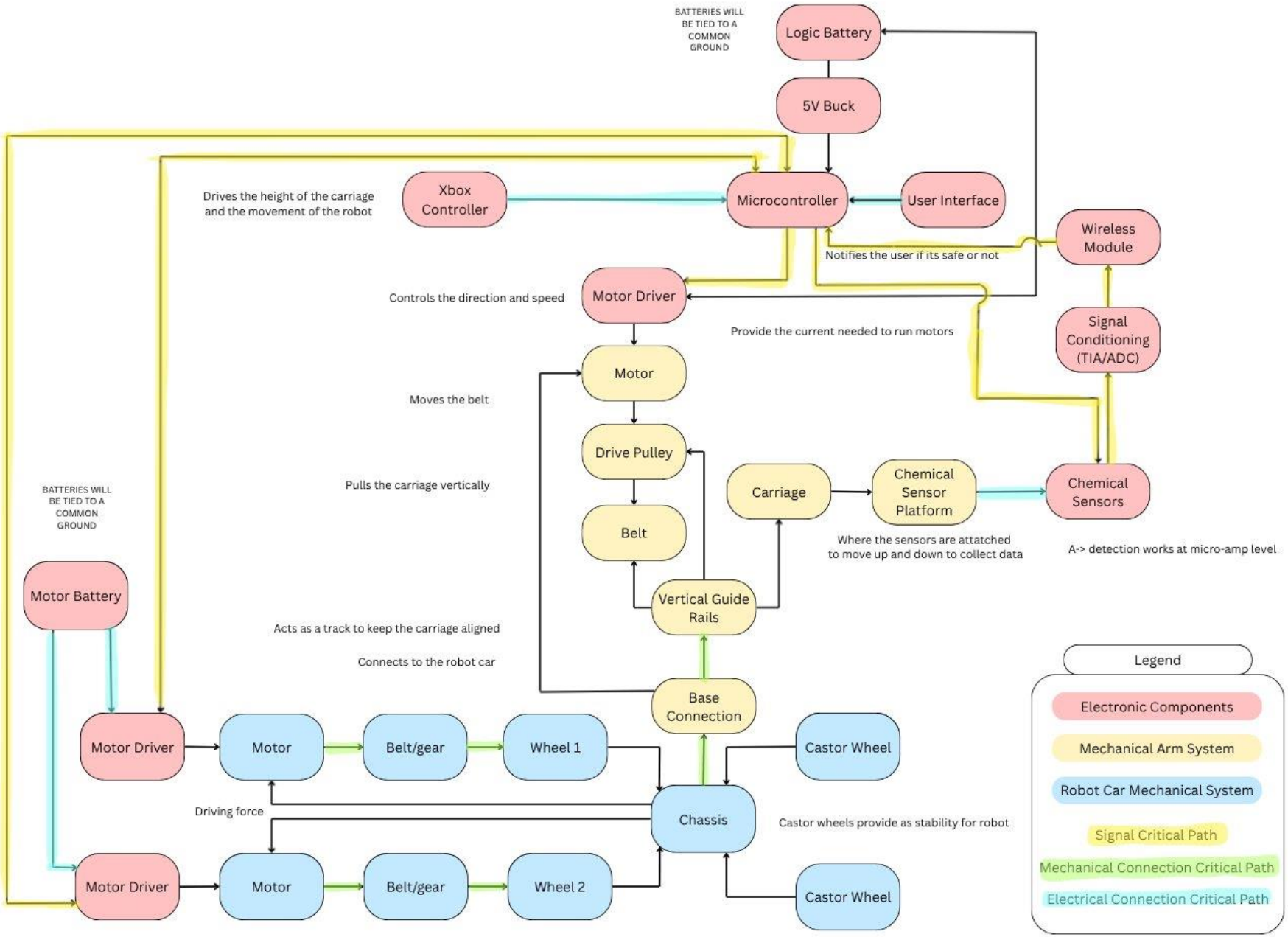
RRP

- CAD a full-scale model for analysis
- Complete calculations for analysis on robot car and arm
- Create an Altium schematic for the power distribution system

Problem Space and Supporting Research

- **Problem Space:** Industrial workers are frequently exposed to dangerous chemical hazards. Over 50,000 annual worker deaths linked to long-term exposure to toxic chemicals and 190,000 are left with illnesses, permanent damage, or diseases like lung disease. (OSHA).

- **Limitations of Existing Market Solutions:**
  - *Fixed Detectors:* High unit cost ($800–$3,000+ each) requiring extensive facility grids, leaving blind spots.
  - *Handheld & Area Monitors :* Personnel must physically enter hazardous zones to deploy or read units, risking direct exposure.
  - *Specialized Mobile Units :* Limited ground-level sampling reach (~dog height) that misses ceiling-pooled gases, high cost, and requires specialized technical training.


BATTERIES WILL BE TIED TO A COMMON GROUND
Logic Battery
5V Buck
Drives the height of the carriage and the movement of the robot
Xbox Controller
Microcontroller
User Interface
Wireless Module
Notifies the user if its safe or not
Controls the direction and speed
Motor Driver
Provide the current needed to run motors
Signal Conditioning (TIA/ADC)
Motor
Moves the belt
Drive Pulley
Pulls the carriage vertically
Carriage
Chemical Sensor Platform
Chemical Sensors
Where the sensors are attatched to move up and down to collect data
A-> detection works at micro-amp level
Belt
BATTERIES WILL BE TIED TO A COMMON GROUND
Motor Battery
Vertical Guide Rails
Acts as a track to keep the carriage aligned
Connects to the robot car
Base Connection
Motor Driver
Motor
Belt/gear
Wheel 1
Castor Wheel
Driving force
Chassis
Castor wheels provide as stability for robot
Motor Driver
Motor
Belt/gear
Wheel 2
Castor Wheel
Legend
Electronic Components
Mechanical Arm System
Robot Car Mechanical System
Signal Critical Path
Mechanical Connection Critical Path
Electrical Connection Critical Path

Critical Path (highlighted paths from block diagram)

Critical Path	Risk	Mitigation
Signal Critical Path 1. Microcontroller to motor drivers 2. Microcontroller to chemical sensors	1. Incorrect control signals could cause the motors to behave unpredictable or not respond. 2. Sensors could send false or unstable readings or fail to communicate with the microcontroller.	1. Check all pin connections and test each motor separately before integrating the full system 2. Verify each sensors operating voltage and output before connections and test each one individually to confirm the microcontroller reads a consistent output.
Mechanical Connection Critical Path 1. Robot chassis to the arm base connection 2. Base connection to the vertical guide rails 3. Motor connection to the belt/gear and wheel	1. The robot car is not designed symmetrically which will be difficult to mount the vertical arm base which could cause tipping and be unstable 2. The vertical arm will consist of 3 stages which will need to be tightly aligned and connect to the belt and carriage without the carriage creating resistance while moving due to its weight. 3. The wheels will move independently of one another and will require two motors which could cause the wheels to rotate at different speeds or veer unexpectedly.	1. We will mount the arm base to the chassis using bolts or screws and a reinforced mounting plate. There will be two rear caster wheels to help maintain stability to avoid tipping with the main wheels acting as control. 2. Use compatible telescoping rails with a small clearance for easy sliding. We will secure the guiderails parallel to each other and use low friction guides or bushings. 3. Each motor will have a separate motor driver to test each motor independently to adjust motor signals compensate for any speed differences.
Electrical Connection Critical Path 1. User interface (feedback) to and from microcontroller 2. Connecting the microcontroller to the Xbox controller 3. Battery wiring to the motors	1. The user interface could display incorrect information if connections are not correct 2. The controller may fail or send delayed or incorrect movement commands. 3. Voltage drops, excessive current draw, loose connections could cause to motors to stop or not respond properly to the controller.	1. Test the outputs and the complete interface to verify commands are displayed correctly 2. Create a timeout or failsafe if the Xbox controller disconnects or a manual stop option. 3. Calculate the current draw of the motors and other components and select batteries with the appropriate current ratings.

Q1 Specifications

Critical Feature List:

1. Drive and adjust to desired testing area
2. Detect harmful chemicals/gases
3. Alert user of dangerous conditions

Spec ID	Requirement	Threshold (Shall)	Objective (Should)	Validation Method	Why this threshold value	Relates to critical feature
RRP001	Sentry Robot Movement Control	Robot shall be able to turn in place and move on flat ground with a 10% overshot	Robot should be able to turn in place and move on an incline/decline of 15 degrees with a 5% overshot	Observe the robots' function on flat ground and on an incline/decline of 15 degrees and measure how accurate turning and distance traveled is.	This demonstrates the robot's ability to function in a typical industrial building. Most warehouses are flat and one level, allowing our robot to be useful in real world situations as long as the threshold value is met.	1
RRP002	Sentry Arm Movement	Arm shall be able to complete a vertical extension of +2 ft within 3 minutes	Arm should be able to complete a vertical extension of +4 ft within 1 minute	Measurement of the arms tip in comparison to its resting (neutral) position	This capability allows the rover to sample air at different heights, including approximately head height, and accommodate environments with varying dimensions.	1
RRP003	Chemical Sensor Detection	The sensors shall be able to detect the presence of their targeted chemical	The sensors should be able to detect if the amount of said chemical is harmful to the people nearby	Unit test on the sensors stacked up against their chemical, then testing a simple code that will alert the user if a certain metric is hit	This allows the rover to detect the presence of a chemical. Measuring concentration rather than just detecting a chemical provides more useful information for assessing potential exposure.	2
RRP004	Alert Indicator to User	The user shall receive notification via the UI that upon detection, the specific chemical is present and harmful.	Within 3 seconds of detection, the Sentry should alert the user via UI which specific chemical is detected, what amount is harmful, and how much is present	Place a sensor under a certainly positive situation, and measure the time it takes from the sensor to communicate its reading to the UI.	This is critical to the user's safety, as the response time can be vital when chemicals that take little time to harm are present. If a user is under the impression that a room is clear because no sensors are tripped, they may enter just to have a late positive test come up in the UI, when the damage could already be done.	3

Q1 Specifications

Critical Feature List:

1. Drive and adjust to desired testing area
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3. Alert user of dangerous conditions

Spec ID	Requirement	Threshold (Shall)	Objective (Should)	Validation Method	Why this threshold value	Relates to critical feature
RRP001	Robot chassis under a load	The robot chassis shall be able to carry the load of the robot arm of 10lbs.	The robot chassis should be able to carry the load of the robot arm of 15lbs.	Design the chassis in SolidWorks and use the dimensions to calculate whether the robot can handle a load of a specified weight using a free body diagram.	This is the threshold value because it proves our design can support the weight a thin-walled aluminum 2-inch tubing arm of 4 posts equally 24 ft (about8 lbs) leaving an extra 2-7 lbs for all the sensors and electrical components.	1
RRP002	Robot centered without tipping	Chassis and arm shall be able to stay centered without tipping at stage 1 of 2ft above the robot chassis.	Arm should be able to stay centered without tipping at stage 3 of 6ft above the robot chassis.	Design the extendable arm in SolidWorks and calculate the center of mass at different heights. Then use that data to calculate if it will tip.	This is the threshold value because it proves our design will be sturdy and not tip while in use in industry settings.	1
RRP003	Chemical sensing signal chain design	In LTspice, the transimpedance amplifier shall convert a simulated sensor current covering the range equivalent to 50–1200 ppm CO into a voltage within the ADC's input range, with error ≤ ±10% of the ideal value across that range	The amplifier should achieve ≤ ±5% error across 0–1200 ppm equivalent, and the ADC → threshold → wireless alert firmware should be demonstrated in a Wokwi simulation.	Model in LTSpice using the op-amp's manufacturer model and a current source standing in for the sensor. Sweep current across the range(derived from datasheet) and compare to ideal output voltage. Calculate the ADC resolution from datasheet to confirm small changes near the 50 ppm threshold are detectable.	No hardware is purchased until winter quarter, so this quarter proves the design on paper and in simulation before buying. 50 ppm is the OSHA PEL for CO and 1200 ppm is the NIOSH IDLH. ±10% leaves margin for the sensor's own tolerance. Simulation results become the baseline the physical prototype is tested against in winter.	2
RRP004	Power Consumption	The electrical system shall account for the estimated power consumption of all electrical	The team should calculate the total estimated power	Draft Altium schematic and run LTspice transient simulations with changing load conditions. Record	Different components require different operating voltages and current levels. A defined power distribution design will	1,2,3

Engineering Analyses

- **Power Consumption:** Total max electrical draw across all components.
- **User Alert Notification:** Calculate the worst-case time from a sensor's output crossing its alarm threshold to the alert appearing at the operator station, using a latency budget:
  - $t_{alert} = t_{ADC} + t_{filter} + t_{proc} + t_{wireless} + t_{display}$
- **Wheel Motor Power:** Force and power needed to drive the robot based on its total weight.
- **Mass Distribution:** Load placement analysis to ensure stability and prevent tipping.
- **Arm Motor Power:** Force required to drive the pulley and carriage system.
- **Structural Analyses:** Force diagrams and stress calculations to ensure the arm and chassis holds under load.

Proposed critical path device

- Electrical (Power consumption)
  - We are designing an electrical system that integrates multiple gas sensors, wireless communication, a microcontroller, drive motors, and an extendable arm actuator. Ensuring that all components receive sufficient and stable power while operating together will pose a challenge. **For the RRP, we propose to research the voltage and current requirements of each electrical component, calculate the total expected power consumption, and select an appropriate battery and power distribution system. This analysis will allow us to establish a reliable electrical design before moving forward with building and testing the refined rover in future quarters.**

- Mechanical (Structural Integrity and Balance)
  - Prior to senior design, we have not constructed a robotic car and arm shaft combination before. The ability to not only construct separate mechanical systems but maintain the structural integrity of it will pose a challenge. **For the RRP we propose to design the basic car and arm using CAD and use several analyses methods including FEA, FBD, COM, and other hand calculations to ensure our design will function as intended and will move forward in future quarters with building our refined robot and arm design.**